Machine Learning-Based Misbehavior Detection in Vehicular Networks with Batch Message Verification

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***Abstract*—The recent advances in wireless communication and onboard sensors have significantly accelerated the development of vehicular networks. The Basic Safety Messages (BSMs) shared among neighboring vehicles through vehicular communications can greatly facilitate traffic management and improve road safety. However, the transmitted BSMs are vulnerable to various attacks like position falsification attack, DoS attack, and Sybil attacks. Many machine learning-based misbehavior detection systems (MDS) have shown great performance in identifying misbehavior messages. While the main stream of ML-based MDS focus on improving the detection performance by designing new features and improve learning model, in this paper we propose a novel MDS that aims at improving the detection efficiency as well as the detection performance in terms of F1-score. The proposed ML-based MDS can verify a batch of BSMs at the same time instead of verifying them one by one so as to improve the detection performance. We have conducted extensive simulations with a public accessible dataset, i.e. Veremi Extension dataset to verify the effectiveness of the proposed MDS. The simulation results shown that our proposed MDS can identify various types of attacks for a batch of messages efficiently with a 98% F-1 score.**

***Index Terms*—Vehicular Networks, Misbehavior Detection, Machine Learning, security, BSMs**

I. Introduction

Vehicular networks are important components of Intelligent Transportation System as well as Internet of Vehicles [1]. Vehicular networks allow vehicles to exchange information with each other, with roadside infrastructure, and with broader networks, enabling applications such as collision avoidance, traffic optimization, and cooperative driving [2]. The ability of vehicles to communicate dynamically improves road safety and efficiency, supporting the development of smart cities and next-generation mobility solutions. A central element of communication in these networks is the Basic Safety Message (BSM) [3]. BSMs convey crucial information about vehicle dynamics, including position, velocity, and acceleration. These messages form the backbone of safety-critical applications, helping to prevent accidents and enhance the flow of traffic [4]. However, the open and distributed nature of vehicular networks makes BSMs susceptible to malicious activities, such as falsified position reports, identity spoofing (e.g., Sybil attacks), and flooding attacks like Denial of Service (DoS) [5]–[7]. Such security risks can disrupt traffic, compromise safety, and undermine the trustworthiness of vehicular communication systems.

Protecting BSMs from various attacks is essential for ensuring the safety and reliability of vehicular networks. Security in vehicular communications is primarily ensured through a Public Key Infrastructure (PKI) [8], which authenticates vehicles and manages digital certificates. However, while these cryptographic measures effectively prevent unauthorized access by external entities, they are less effective against insider threats, where authenticated entities act maliciously [9], [10]. To address this limitation, Misbehavior Detection Systems (MDSs) have emerged as a promising solution [11], [12]. These systems aim to detect and mitigate potentially harmful activities within the network by analyzing V2X message data and identifying misbehaving entities, thereby enhancing the overall security and trustworthiness of V2X applications.

Recently, artificial intelligence, particularly machine learning (ML) and deep learning approaches, has gained significant attention in enhancing misbehavior detection in vehicular communications. While conventional MDSs rely on predefined semantic rules, such as "acceptance range threshold," "minimum distance moved," and "sudden-appearance warning," ML-based MDSs offer a more dynamic and adaptive approach. By leveraging ML algorithms, these systems can detect and classify anomalies in vehicular communication with higher accuracy and F1-score. The availability of publicly accessible datasets specifically designed for vehicular communication, such as the Vehicular Reference Misbehavior (Veremi) dataset [13] and its Veremi Extension [14], has further accelerated advancements in this field. Since 2018, the number of ML-based MDSs has grown rapidly, with

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studies demonstrating significant improvements in detection performance compared to traditional methods [15].

Despite the notable success of ML-based MDS in improving detection accuracy, many MDSs are optimized to process messages individually. This approach can be computationally intensive and inefficient, particularly in high-traffic communication scenarios, posing significant challenges for real-time deployment. To address these limitations, this paper introduces a novel batch verification approach that processes BSMs simultaneously. This method enhances detection efficiency without compromising accuracy, making it well-suited for scalable, real-time applications in vehicular networks. The key contributions of this work are as follows:

- We present a new ML-based MDS deployed on vehicles capable of accurately detecting 19 types of attacks.
- The proposed architecture significantly reduces computational overhead by verifying multiple BSMs in batches, ensuring suitability for high-density vehicular scenarios and safety-critical applications. By prioritizing efficiency alongside detection performance, this work introduces a novel dimension to MDS design, offering practical solutions to real-world vehicular network security challenges.
- Comprehensive simulations using the publicly available Veremi Extension dataset validate the system's effectiveness. Performance evaluations demonstrate that the proposed MDS outperform tradition ML methods in terms of F1-score and computational efficiency.

The remainder of this paper is organized as follows. Section II reviews existing ML-based MDS for vehicular communications. Section III details the proposed MDS architecture and data processing methods. Section IV presents the simulation environment and numerical results. Finally, Section V concludes the paper.

II. Related Work

Securing vehicular communications is a challenging task. Many works have been proposed for misbehavior detection in vehicular networks to mitigate threats from insiders. Generally, those MDS can be categorized into traditional MDS and ML-based MDS. In [13], the proposed framework, Maat, employs subjective logic to establish a fusion and data management system for assessing data trustworthiness. The authors evaluated its performance using four methods: Acceptance Range Threshold, Sudden Appearance Warning, Simple Speed Check, and Distance Moved Verifier. The VeReMi dataset, generated through simulations, was used for testing. Zaidi *et al.* [16] developed an MDS capable of identifying fake information attacks and other threats using statistical methods, without relying on trust or reputation scores. The approach assumes that neighboring vehicles typically experience similar traffic flow levels. If a vehicle reports a flow value inconsistent with others, the data is discarded, and the vehicle's ID is flagged. So *et al.* [17] addressed the issue of inadequate compartmentalization in vehicles by examining the Received Signal Strength Indicator (RSSI). Their work focused on the physical layer, utilizing a plausibility model to detect falsified node locations. The detection was performed in each vehicle locally. Kamel *et al.* [14] compared the performance of Threshold based method, Non-Cooperative Trust based method, Cooperative Trust based method, and ML-based methods for misbehavior detection with their created Veremi Extension dataset. Their testing results showed that ML-based methods outperforms the traditional (non-ML) methods in terms of F-1 score. With the simulated datasets for vehicular communications [13], [14] became publicly available, many ML-based MDS are proposed and investigated.

Gyawali *et al.* proposed a machine learning-based MDS to identify false alert and position falsification attacks [18]. For false alert detection, they trained a binary classifier that analyzed discrepancies between the average traffic flow and the flow values reported by vehicles, calculated using vehicle density and average speed. They tested the performance of the binary classifier with their own simulated dataset. To address position falsification attacks, they employed a multi-class classifier trained on the VeReMi dataset, utilizing features such as speed and position changes between consecutive beacons, receiving distance, RSSI, and vehicle dynamics.

Alladi *et al.* proposed a mobile edge computing (MEC) based MDS to identify various types of attacks in vehicular networks [19]. In [19], two misbehavior detection strategies were proposed and investigated. In the first method, the collected messages with time sequence were directly sent into a CNN-LSTM model, which comporised of two CNN layers and one LSTM layer. The second method converted 20 sequential messages into a image and then used CNN to identify misbehaviors. They used the Veremi+ dataset for simulation and showed that the image based MDS achieves better performance with a 99.65% accuracy and 98.26% F-1 score when identifying 9 different types of attacks. Simulation results also showed that the image based classification with CNN can be completed in 14.29 ms using Rasberry Pi 3B device, justifying its efficiency in realistic scenario. Besides [19], T. Alladi *et al.* investigated in another MDS that is based on a data reconstruction method [20]. The received consecutive messages were treated as input sequence that went through a CNN-LSTM reconstruction model to generate a reconstructed sequence. If the Mean Absolute Error between the input sequence and the output sequence exceeds a threshold, then the input sequence would be identified as malicious.

Ercan *et al.* introduced three new features, i.e. estimated angle of arrival, estimated distance between sender and receiver, and the differences between the declared and estimated distance between sender and receiver, to identify position falsification attacks [21]. They implemented KNN, RF, and an ensemble learning that combines KNN and RF to exploit the potential of ensemble learning. Their extensive simulation based on the Veremi dataset have shown that the ensemble learning performed better in low vehicle density scenario and the RF worked better in medium and high density scenario.

Sharma and A. Jaekel explored a MDS that combined information of two consecutive BSMs to identify position falsification attacks [22]. The position and speed information

TABLE I: Comparison of Related Works

Work	Deployment	Dataset Used	Detection Methods	Batch Verification
Maat *et al.* [13]	Locally	VeReMi	Subjective logic	No
Zaidi *et al.* [16]	Locally	Simulated	Statistical methods	No
So *et al.* [17]	Locally	Not specified	Plausibility model (RSSI-based)	No
Kamel *et al.* [14]	Locally	Veremi Extension	ML-based and traditional methods	No
Gyawali *et al.* [18]	Locally	VeReMi, custom	Binary and multi-class classifiers	No
Alladi *et al.* [19]	Locally	Veremi+	CNN-LSTM, CNN (image-based)	No
Alladi *et al.* [20]	Locally	Not specified	CNN-LSTM (reconstruction model)	No
Ercan *et al.* [21]	Locally	VeReMi	KNN, RF, ensemble learning	No
Sharma *et al.* [22]	Locally	VeReMi	KNN, Random Forest, Naive Bayes, Decision Tree	No
Huang *et al.* [23]	Locally	VeReMi	Multi-objective ML with SVM	No
Sedar *et al.* [24]	Globally	Veremi+	Reinforcement Learning (Q-network with LSTM)	No
Valentini *et al.* [25]	Locally	NS3 and SUMO	Median Absolute Deviation	No
Lv *et al.* [26]	Globally	Not specified	Geographic info, social relationship, reputation	No
Ours	Locally	VeReMi+	CNN based batch verification	Yes

from two consecutive BSMs from the same sender were combined into one record. The authors used the Veremi dataset to create a new dataset where two BSMs are combined into one. Then, they compared the detection performance of KNN, Ramdom Forest, Naive Bayes, and Decision Tree. Their simulation results showed that using two BSMs outperforms the one BSM based method, and the KNN was the best among the compared models.

Huang *et al.* explored a multi-objective machine learning model designed to enhance detection performance in terms of the F1-score [23]. During the training phase, the model automatically selected the most relevant features from the input data and identified a set of Pareto-optimal models that simultaneously maximized recall and specificity. For misbehavior detection, each input message was evaluated by the selected Pareto-optimal models, and their weighted outputs were combined to produce the final decision. Using the Veremi dataset for evaluation, the simulation results demonstrated that their proposed model outperformed both [13] and [22].

Sedar *et al.* investigated a reinforcement learning (RL)-based MDS [24]. In [24], the MDS was deployed at RSUs, where the V2X time-series data can be gathered and the device has superior computational resource. The agent took the received BSMs and prior system state information as input into a Q-network, which was composed by an LSTM layer with a fully-connected neural network. The action space has two values, where the value 1 indicated the detection of an attack and value 0 presented a normal state. They used the Veremi+ dataset for simulation and exhibited that the RL-based MDS can detect all types of attacks in the Veremi+ dataset with a high F-1 score. However, the computation complexity was not discussed.

Valentini *et al.* proposed a decentralized can cooperative MDS named DAMASCO to detect DoS/DDoS attacks [25]. They utilized the information from Medium Access Control sublayer to calculate a Median Absolute Deviation value to distinguish attacks and normal behaviors. Simulations based on Network Simulator 3 and SUMO have shown the DAMASCO achieved 95% detection rate with 3% false positive rate.

Lv *et al.* proposed to utilize geographic information and social relationship of vehicles to identify fake message attack, gray hole attack, bad-mouth attack, and time-shift attack [26]. The spatial and time information in the received message was used to measure the message trust, and the similarities in geographic movement patterns of vehicles were used to construct social relationships. A reputation system combined the information of message trust and social relationships to identify misbehavior. This MDS can achieve a 96.33% detection rate based on their simulation results. However, the time consumption has not been discussed. Whether this method is effective for delay-sensitive applications remain questionable.

Though many ML-based MDS have been proposed and exhibited excellent performance in identifying misbehavior, few of them concerned about efficiency in their design. ML models can be complicated and require considerable computational resource for vehicle grade processors. The MDS that verifies one message at a time may introduce significant delay in high density scenario. Thus, in this paper we propose a novel MDS architecture that can verify messages in a batch to improve the detection efficiency with general ML algorithms like CNN.

III. The Proposed MDS

A. The system model

In this work, we consider a general vehicular communication network where vehicles broadcast BSMs to surrounding vehicles at fixed intervals. The communication can be based on either DSRC or C-V2X protocols. The MDS is directly deployed on OBUs to minimize detection delay, allowing vehicles to receive results promptly and respond immediately. Each BSM contains kinematic information (e.g., position, speed, steering) and identity details (e.g., pseudoID, timestamp), along with a digital signature. The digital signature ensures message authentication, preventing unauthorized entities from joining the network. While a valid signature guarantees the BSM was sent by a legitimate source and remains unaltered during transmission, it cannot verify the accuracy or trustworthiness of the numeric values within the message. The primary objective of the MDS is to identify misleading BSMs originating from insiders, whether due to faulty sensors or malicious tampering.

In this work we only consider attacks from insiders and misbehavior are lunched by insiders who posses legitimate identity to pass message authentication process. The misbehaved vehicles send abnormal BSMs that contains false or fabricated information. Specifically, we consider the following attacks that may occur in the vehicular networks through contaminated BSMs:

- **Position Falsification Attack**: The attacker broadcast a faulty position information to let the receiver believe the sender is at the stated position. For example, the attacker can continuous broadcast a fixed position, or a random position, or add a fixed/random offset on its real position.
- **Speed Falsification Attack**: The attacker broadcast a faulty speed information, e.g. a fixed speed, a random speed, and add a fixed/random value on its real speed.
- **DoS Attack**: The attackers may launch a DoS attack by sending BSMs in a frequency that exceeds the standard. The large volume of transmitted BSMs may overwhelm the communication channel of vehicular networks.
- **Disruptive Attack**: The attacker utilizes received genuine BSMs to forge a fabricated BSM, which is plausible and is more likely to bypass the detection.
- **Relay Attack**: The attacker send BSMs that contains same information as previous ones.
- **Sybil Attack**: The attacker use different pseudoID for BSMs sent in a short time perioud. Usually it is combined with other types of attacks like DoS attack and Disruptive attacks.

All the received BSMs need to be verified by passing through a MDS in the vehicle to mitigate misbehavior.

B. Misbehavior Detection

In this work, we propose a novel MDS architecture that converts sequential BSMs to a three-dimension data unit (3DU) and then use a machine learning model to classify the message as normal or abnormal. For the ML model, in this work we mainly consider the CNN since it is known for its excellent capacity in handling 3D data like images. Different from other image based MDS that convert message features into image pixels, we treat each message as a whole to fit into a pixel in the 3DU. Here, we will introduce three different methods to construct different 3DU for misbehavior detection. All the three methods share the same MDS architecture but convert the BSMs into different format.

The architecture of the proposed MDS is shown in Fig. 1. First, the OBU preprocesses the received BSMs to extract necessary features, e.g. position, speed, and acceleration information. Each vehicle has a local buffer to store BSMs that sent by the same sender and then convert them into the designed 3DU format. Then the converted 3DU is sent to a pre-trained CNN model to identify the abnormal BSM.

Single Message Verification: Assume the OBU can buffer up to n BSMs associated with the same pseudoID. When the OBU receives a BSM from a new sender with pseudoID$_i$ (denoted as sender P_i and message M_{ij}), it initializes a 3D data structure (3DU) of dimensions $1 \times n \times k$, where

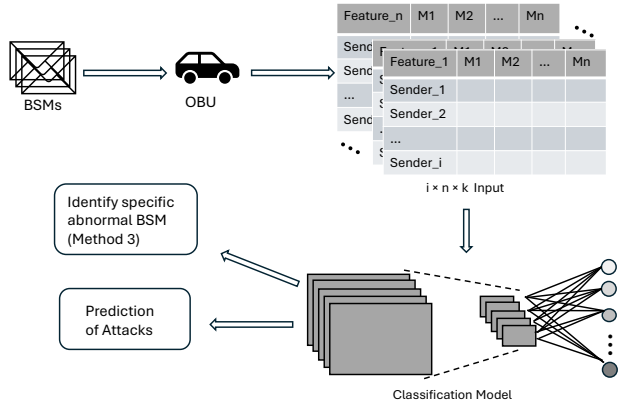


Fig. 1: Proposed MDS structures

k represents the number of selected features relevant for misbehavior detection. The data in M_{ij} is placed in $(1, n, :)$, while all other positions in the 3DU are initialized to zeros.

Upon receiving a second BSM from P_i, denoted as M_{i2}, the existing data in $(1, n, :)$ is shifted one position to the left, occupying $(1, n-1, :)$, and the new data from M_{i2} is written into $(1, n, :)$. This process repeats as the OBU continues to receive BSMs from P_i. When the j-th message from P_i is received ($j \leq n$), the data of M_{i1} is pushed to $(1, n+1-j, :)$, and the latest data, P_{ij}, is stored at $(1, n, :)$. For $j > n$, only the most recent n messages are retained in the buffer, ensuring the 3DU always contains data from the latest n sequential BSMs with the same pseudoID.

Every time the OBU receives a new BSM, it updates the corresponding 3DU with the incoming data and uses a pre-trained CNN model to analyze the structure. The model determines whether the current 3DU indicates any misbehavior. The above procedures are summarized in Algorithm I.

Algorithm 1 Single Message Verification for MDS in Vehicular Networks

Input: BSM M_{ij} from with pseudoID$_i$
Output: Detection result of misbehavior for sender P_i

```
if pseudoID_i is new then
    Initialize 3D data structure 3DU of size 1 × n × k with zeros
    Write data from M_ij into 3DU(1, n, :)
else
    for each new M_ij received do
        Shift existing data in 3DU left by one position:
        3DU(1, l, :) ← 3DU(1, l + 1, :), ∀l = 1 to n − 1
        Write data from M_ij into 3DU(1, n, :)
    end for
end if
Feed 3DU into the pre-trained CNN model
Return detection result for M_ij
```

Algorithm 2 Batch Message Verification for MDS in Vehicular Networks

Input: Batch of BSMs, M_{ij}, from multiple senders P_i with unique pseudoID$_i$

Output: Detection result of misbehavior for the batch and misbehavior index

Initialize a 3D data structure (3DU) of dimensions $I \times n \times k$, filled with zeros
for each received BSM M_{ij} from sender P_i **do**
  if $i \leq I$ **then**
    Shift all data in row i one position to the left:
    $(i, l, :) \leftarrow (i, l+1, :), \forall l = 1$ to $n-1$
    Place the data of M_{ij} in $(i, n, :)$
  else
    Allocate a new 3DU for additional BSMs **or** switch to single message verification for M_{ij}
  end if
end for
for each updated 3DU **do**
  Use the first CNN model to classify the 3DU as normal or a specific attack type
  Use the second CNN model to identify the misbehavior index (problematic data within the 3DU)
end for
return Detection results for the batch and misbehavior index

Batch Message Verification (BMV): As vehicle density increases, the OBU may receive a large number of BSMs within a short time. Verifying all received BSMs with minimal delay is crucial for maintaining road safety. Here, we describe a data construction method to enable batch message verification. Assume the OBU can buffer up to n BSMs for each unique pseudoID, and the batch size is I. The OBU initializes a 3D data structure (3DU) of dimensions $I \times n \times k$, filled with zeros. When the OBU receives a message M_{ij} from sender P_i, all data in the i-th row of the 3DU is shifted one position to the left, i.e., $(i, l, :) \leftarrow (i, l+1, :)$ for $l = 1$ to $n-1$. The new data from M_{ij} is then placed in $(i, n, :)$.

If $i > I$, meaning the number of unique pseudoIDs exceeds the batch size, the OBU must either allocate a new 3DU for additional BSMs or revert to the single message verification structure to individually verify the extra BSMs.

Unlike single message verification, batch message verification must identify not only whether there is misbehavior within the batch but also which specific message(s) in the batch are problematic. To achieve this, two CNN models are employed to analyze the updated 3DU. The first model classifies the entire 3DU as either normal or a specific attack type, similar to the approach used in single message verification. The second model identifies the exact data within the 3DU that is abnormal, which is referred to as the "misbehavior index" in this paper. The description of batch message verification is summarized in Alogirithm 2.

TABLE II

Label	Type	Label	Type
0	Normal	10	Disruptive
1	ConstPos	11	DataReply
2	ConstPosOffset	12	DelayedMessages
3	RandomPos	13	DoS
4	RandomPosOffset	14	DoSRandom
5	ConstSpeed	15	DoSDisruptive
6	ConstSpeedOffset	16	GridSybil
7	RandomSpeed	17	DataReplySybil
8	RandomSpeedOffset	18	DoSRandomSybil
9	EventualStop	19	DoSDisruptiveSybil

TABLE III: Confusion Matrix

	Predicted normal	Predicted malicious
Actual normal	TN	FP
Actual malicious	FN	TP

IV. Simulation Results and Performance Analysis

In this section, we describe the simulation environment, present the results, and provide a discussion of the findings along with key insights derived from the simulations.

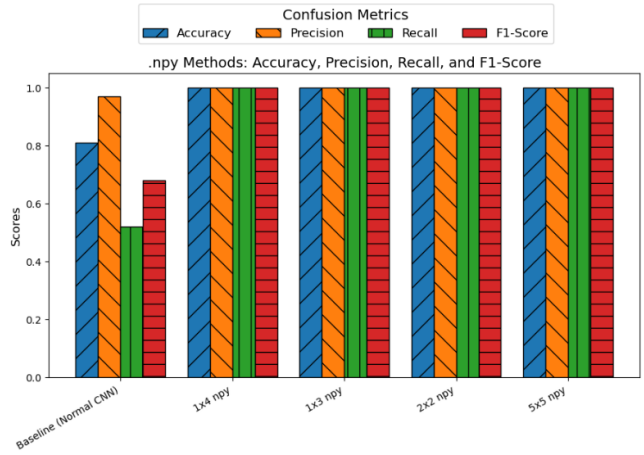


Fig. 2: Binary classification 1xN accuracy

A. Simulation Setup

In order to validate the performance of the proposed MDS and provide a comparable results with other works, we use the VeReMi+ [14] dataset for simulation, which is public available and widely adopted. The VeReMi+ dataset was created with Framework For Misbehavior Detection [27], an extension of Vehicles in Network Simulation (VEINS) simulator [28]. VEINS, an open-source framework for vehicular network simulations, integrates two key tools: Objective Modular Network Testbed in C++ (OMNeT++) [29] for network simulation and the Simulation of Urban MObility (SUMO) [30] for road traffic simulation. The original dataset includes log files capturing traces from every vehicle that traversed the network over a 24-hour period as well as a ground truth file for each hour of the simulation, which provides data points for

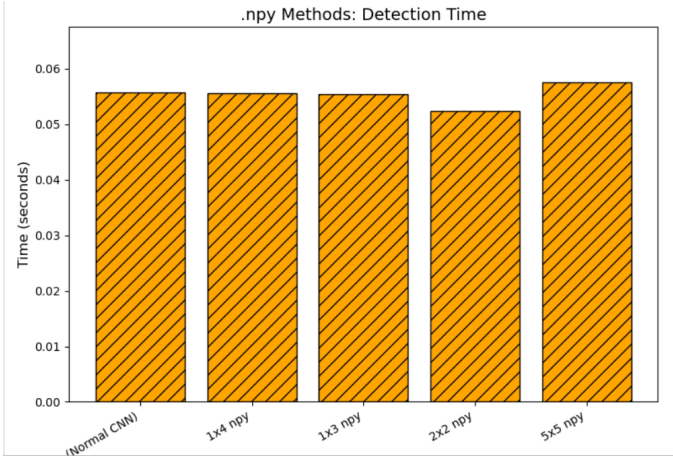


Fig. 3: Binary classification 1xN prediction time

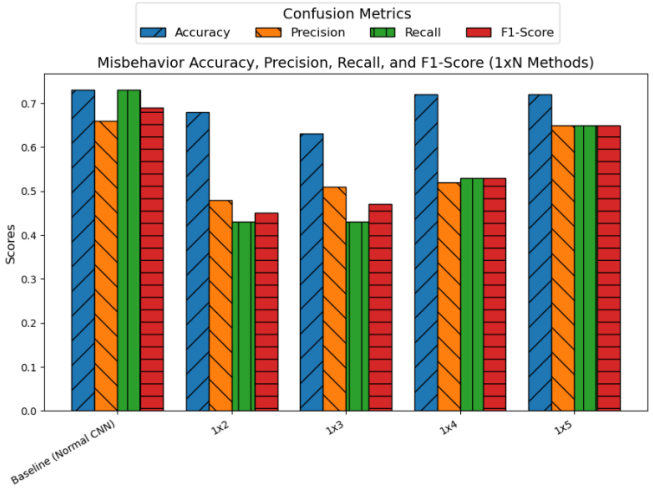


Fig. 4: Multi-class 1xN classification accuracy

all vehicles that traveled the network during that hour. Each data point represents a message transmitted by a vehicle in motion within the network and comprises various fields. There are 19 attacks simulated in the VeReMi+ dataset. Here we present all the attacks in Table II , and detailed description of attacks can be found in [14]. To simplify the data processing procedure, we used the pre-processed VeReMi+ dataset [31] that contains 3194808 instances with 30 features in CSV format. After obersving all features in pre-processed VeReMi Entension dataset, we found many features only contain zeroes, which are meaningless for MDS. So we selected 12 features out of 30 for training and testing. The selected features are "sendTime", "senderPseudo", "posx", "posy", "posx_n", "posy_n","spdx_n", "spdy_n", "aclx_n", "acly_n", "hedx_n", "hedy_n". Before we start our simulation, we need to create many 3DU dataset with different dimensions. The 3DU dataset generation process involves converting pre-processed data into $i \times n$ pixel images with 12 channels, where each channel represents a normalized numeric feature from the dataset. The process begins by splitting the data into normal and attack classes. Normal images are generated by sampling groups of n consecutive BSMs from i senders the normal class, sorting them by timestamp, and populating the image grid row by row. Each cell in the grid corresponds to a data record, and its pixel intensity is computed by scaling the feature values to a range of 0–255. For images with abnormal messages, BSMs from normal and attack classes are combined, and a misbehavior index is introduced to identify where the attack data appears in the image. We use a numeric value to present the misbehavior index. For example, in a 5×5 image with the abnormal message located at the second row and the fifth column, then the misbehavior index is 10. Special handling ensures rows are repeated if insufficient data is available and highlights misbehavior cells with distinct pixel intensities.

B. Simulation Results

We conducted extensive simulations to show the excellent performance of the proposed MDS. To assess the effectiveness of the proposed framework, we utilize precision, recall, and F1-score as performance metrics. These metrics are computed using Equations (1)–(4), based on the confusion matrix defined in Table II.

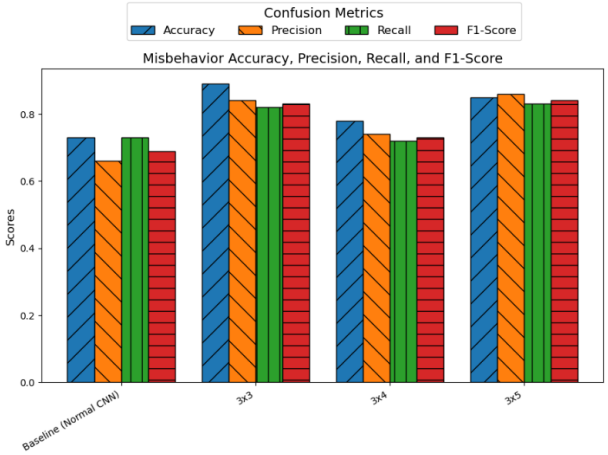
$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{4}$$

First we evaluated the performance of the proposed methods in binary classification, i.e. only distinguish the message as normal or abnormal. To show the effectiveness of our proposed MDS structures, we use the CNN with original tabular BSM data as baseline. Our proposed MDS is labeled as "$i \times nnpy$", where i presents the number of messages being detected at a time, n is the number of consecutive messages received from the same sender, and we saved the created 3DU as "$.npy$" file. For the binary classification, our proposed method achieves excellent performance for different selections of i and n. The Fig.2 shows the results for 1×3, 1×4, 2×2, and 5×5 frame size, and all of them achieve 100% accuracy, precision, recall, and F-1 score. The baseline CNN method can only achieve 0.8 accuracy, 0.96 precision, 0.52 recall, and 0.68 F-1 score. As we can see from the figure, our proposed MDS method significantly improves the detection performance. The Fig.3 presents the average detection time for each detection method, which is measured by averaging the time of testing 500 samples. From the Fig.3 we can see that the detection time only sightly increased as we increase the dimension of the data frame. Although the detection time for $5 \times 5npy$ is slightly higher than the baseline CNN method, it tests BSMs from 5



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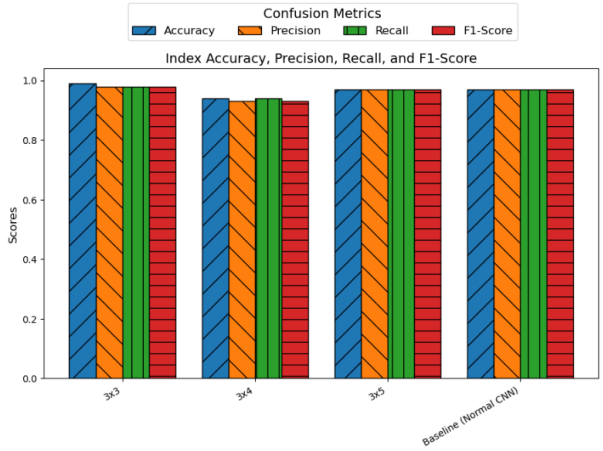
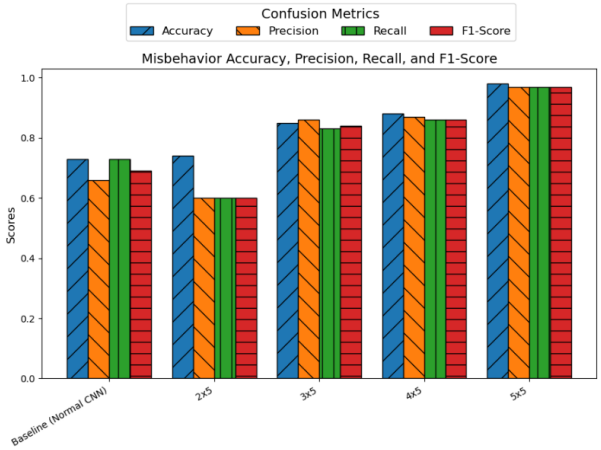
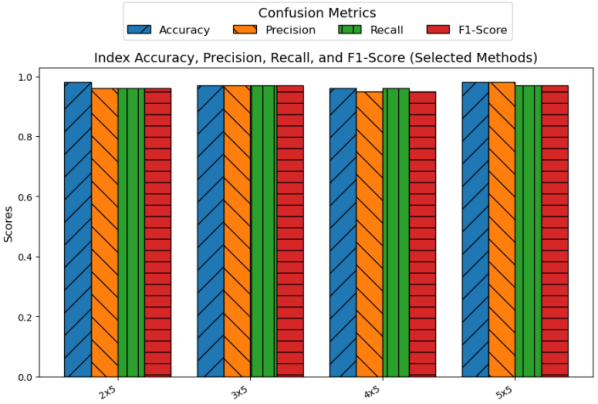


(b)

Fig. 5: An overall caption for the figure, with subfigures labeled (a) and (b).



(a)



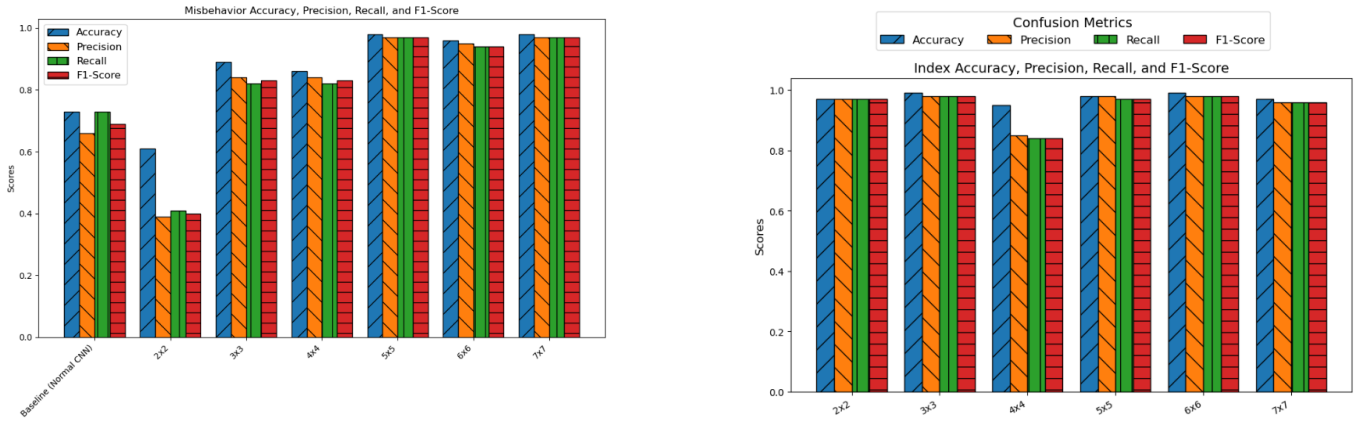
(b)

Fig. 6: An overall caption for the figure, with subfigures labeled (a) and (b).

senders at once, which reduced the computational overhead to one fifth of the baseline CNN method.

The multi-class verification will classify the BSM into 20 classes (1 normal class and 19 attacks), which is much more challenging than binary classification. We first explore the performance of multi-class verification in single message verification settings. The Fig.4 presents the results of the Single Message Verification method with i sets to 1, and n sets from 2 to 5. We still consider a CNN model applied to the original tabular BSM data as the baseline model for comparison. From the figure we can see that the detection performance improves as the n increases. The larger the number of n means more historical data from the same sender can be utilized, which increase the detection performance. The performance of the proposed MDS with 1×5 framework is slightly lower than the baseline CNN method. However, based on the trend presented in the figure, we can infer that the proposed MDS will outperform the baseline model as we keep increasing n. As for the computational overhead shown in Fig.8a, the detection time for single message verification are very close for all simulated frame size, which is around 0.06s.

For Batch Message Verifications, we tested the performance with different selection of i and n and evaluated both the multi-class classification performance as well as the index detection performance. The multi-class classification present how well the proposed BMV classifies 20 types of BSMs. The index detection aims at identifying the location of the abnormal message within the $i \times n$ data frame so that we can know which abnormal message should be discarded instead of all the messages in the data frame. For multi-class classification, we still use the CNN model applied to normal tabular BSM data as the baseline method for comparison. For the index detection, since it is specific to our proposed misbehavior detection data frame, there is no baseline method for comparison. And the index detection is only needed under batch message



(a) (b)

Fig. 7: An overall caption for the figure, with subfigures labeled (a) and (b).

verification Scenario, so we didn't discuss it in the previous single message verification scenario.

Fig.5 presents the simulation results as we fix $i = 3$ and set $n \in \{3, 4, 5\}$ to explore the relation between the detection performance and values of n. The fig.5a presents the results of multi-class classification and the fig.5b presents the results of index detection. From the fig.5a we can see the 3×5 frame size has the highest F-1 score, which is around 15% higher than the baseline CNN model. However, the F-1 score of 3×5 frame size is only around 0.82, which has large room for improvement. The fig.5b shows that all the simulated methods achieve excellent results in index identification. The 3×3 has the best index detection performance with 0.99 accuracy, 0.98 precision, 0.98 recall, and 0.98 F-1 Score. The Fig.8b presents the computation overhead for multi-class classification. The 3×5 frame size takes almost the same time as the normal CNN and slightly lower than 3×3 frame size. Since the $3 \times N$ frame size can verify messages from three senders at once, which reduces the detection time to one third of the baseline CNN method.

The Fig.6 compares the detection performance for different values of i while setting $n = 5$. The Fig.6a shows the results of multi-class classification and Fig.6b shows the results of index detection. From Fig.6a we can see that as the value i increases, the detection performance increases as well. The larger i means we use information from more senders to detect misbehavior which might be helpful in identifying DoS and Sybil attacks. When $i >= 3$, the proposed MDS method outperforms the baseline CNN method. When the data frame size is set to 5×5, it achieves 0.98 accuracy, 0.97 precision, 0.97 recall, and 0.97 F-1 score. The Fig.6b presents the index detection results, from which we can see that the index detection performance are all close to 0.98 and not affected by selection of is. The classification delay is shown in Fig.8c. From the figure we can see the delay of the proposed MDS is very close to the baseline CNN method for one testing. We also observe that the detection delay does not increase as we increase i.

To further explore the impact of different values of i and n to the detection performance, we conducted simulations where we increased i and n simultaneously. From Fig.7a we can clearly see the trend that the detection performance increases as we increasing the dimension of the data frame. When the data frame size is larger than 5×5, the detection performance only changes slightly. The Fig.7b shows that the index detection performance are all very good except the 4×4 frame size. The index detection performance of 7×7 is slightly lower than the 5×5 cases. This may due to the number of messages in the 7×7 frame size is almost doubled compared to the 5×5 frame size. As for the multi-class classification delay, the Fig.8d further proves that the increase in data frame dimension does not introduce extra delay for the proposed method. As the number of messages increases in our data frames, the advantage of our proposed batch message verification delay becomes more significant.

In summary, our proposed MDS framework significantly improves the detection performance in terms of accuracy, precision, recall, and F-1 socre in both binary classification and multi-class classification compare to the traditional CNN method. Moreover, our proposed MDS supports batch message verifications which significantly reduces the computational overhead by verifying multiple messages simultaneously. The excellent misbehavior detection performance and the low latency are critical for secure vehicualr communications.

V. Conclusion and Future Work

This paper proposes a novel MDS capable of identifying and classifying 19 distinct types of attacks in vehicular networks. Unlike existing methods, our approach achieves batch message verification within the same timeframe that other machine learning-based MDSs require to verify a single message, all while maintaining exceptional detection performance. The proposed method aggregates newly received BSMs from multiple senders with buffered BSMs to construct a 3DU. This 3DU is then processed by two machine learning models to classify the

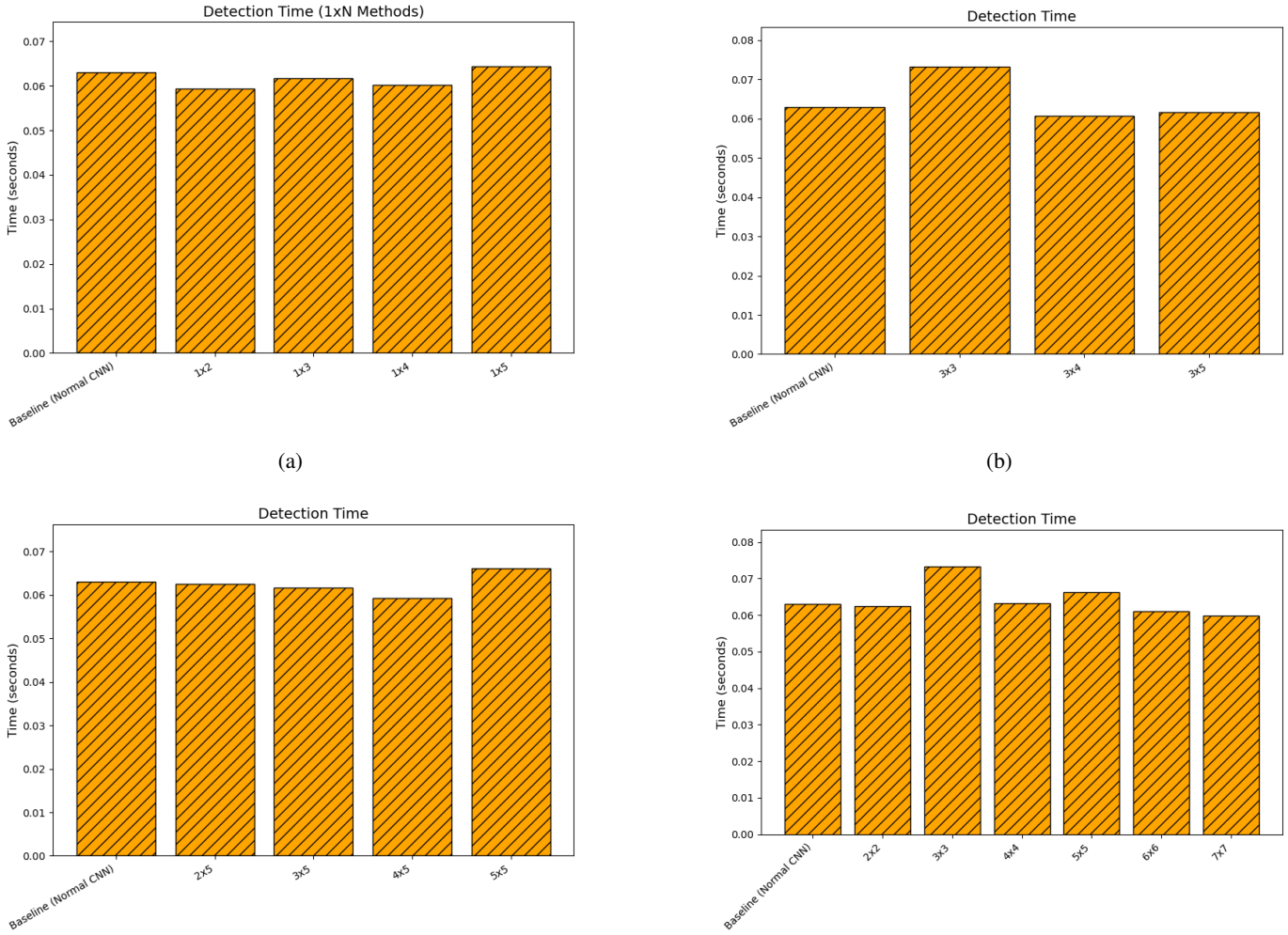


Fig. 8: An overall caption describing the four subfigures, labeled (a), (b), (c), and (d).

misbehavior type and pinpoint the location of the malicious message within the 3DU. We conducted simulations using the Veremi+ dataset, exploring performance across varying 3DU dimensions. The results demonstrate that our method achieves a 100% F1-score in binary classification, regardless of the 3DU dimensions. For multi-class classification across 20 classes, the proposed method delivers excellent detection performance when the 3DU dimension exceeds Theoretically, there is no upper limit on the number of senders (i.e., the number of rows) contributing to the 3DU, and increasing the number of rows further enhances computational efficiency compared to non-batch verification methods. In future work, we plan to evaluate the system's ability to handle mixed attack types within a single 3DU, assessing whether it can simultaneously identify multiple attacks from multiple senders. Additionally, we will test our model in real-world scenarios using actual V2X communication equipment.

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